

11.2 Evaluating Expressions With Exponents

Lesson Introduction

 Benchmark	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.		 Learning Target	I can evaluate expressions that include exponents.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		MA.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.	
 Essential Questions	- How do you evaluate an expression when it includes an exponent? - What information is needed in order to “evaluate” an algebraic expression?			
 Lesson Narrative	Lesson 2 targets the inclusion of exponents in evaluating expressions. Students were introduced to evaluating exponents in Unit 7, but this lesson will be the first time they are evaluating exponents within numerical and algebraic expressions that include multiple operations. The lesson begins with students analyzing one of the questions from Lesson 1 for errors. The 11.2.2 Collaboration and 11.2.3 Guided Instruction then incorporate exponents formally for students to assimilate and make sense of expressions with exponents.			
 Component Summary	11.2.1 Warm-Up (~5 min.)	Analyze student work for errors (<i>SE p. 245</i>)	11.2.4 Try It (10 min.)	Evaluate algebraic expressions with exponents (<i>SE p. 248</i>)
	11.2.2 Collaboration (10 min.)	Sequence the steps in the order of operations and explore how exponents can be written out as products (<i>SE p. 245</i>)	11.2.5 Your Turn! (10 min.)	Evaluate numeric and algebraic expressions with exponents (<i>SE p. 250</i>)
	11.2.3 Guided Instruction (15 min.)	Evaluate numeric expressions with exponents (<i>SE p. 247</i>)	11.2.6 Wrap-Up (~5 min.)	Find and explain errors in students’ work (<i>SE p. 251</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Order of operations <u>Additional Terms:</u> N/A		(optional) Colored pencils	
			Additional Preparation	
			If splitting the class into groups, consider reviewing assessment data from the previous lesson to determine groupings.	

 Teacher Tips	Common Misconceptions		Things to Consider
	• Avoid using the mnemonic PEMDAS to help students remember the order of operations, because three issues may arise with the application of this mnemonic. ○ Students may incorrectly believe that they should always do multiplication before division and addition before subtraction. ○ There is flexibility within the order in certain situations. For example, in the expression $32 - 4(2 + 7) + 8 \div 4$, students have options as to where they might start. In this case, they may first add $2 + 7$ in the grouping symbol or divide $8 \div 4$ before doing any other computation, all without affecting the outcome. ○ The P in PEMDAS suggests that parentheses are first rather than grouping symbols more generally, which would include brackets, braces, square root symbols, and the horizontal fraction bar.		• The lesson includes some historical information on order of operations and why a “rule” became necessary when programming computers to perform operations. Consider incorporating more information on the history of computers as devices designed to “compute values” and how today computers are used for more than computing values. • If time is an issue, then consider assigning 11.2.5 Your Turn! for homework.
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect For 11.2.4 Try It, consider using the Compare and Connect routine by encouraging students to compare how the values of the expressions change depending on the order in which the operations are performed.		MA.K12.MTR.5.1: Patterns and Structures This lesson builds upon student learning from Lesson 1. Students apply what they have learned about the order of operations to expressions that include exponents, extending their knowledge to evaluate more complex expressions.
 Differentiate Instruction	For 11.2.4 Try It, consider placing students in differentiated groups during this activity. Students who have demonstrated a good grasp of the content can be assigned to work independently to complete the questions, while students who may still be struggling can work through the content with guidance from the teacher before trying independently.		
Lesson Notes:			

11.2.1 Warm-Up: Error Analysis

Component Overview: Students analyze two pieces of student work to determine the errors. To highlight specific misconceptions made by your class, consider using common errors from students' exit tickets from Lesson 1. Alternatively, use the common mistakes from students' work in the lesson.



11.2.1 Warm-Up: Error Analysis

Two students made errors on yesterday's exit ticket. Their work is shown below.

Evaluate $8a + 3b(2a - 4)$ for $a = 2$ and $b = 6$.

Azalea's Work	Joshua's Work
$82 + 36(22 - 4)$	$8(2) + 3(6)[2(2) - 4]$
$82 + 36(18)$	$16 + 18(4 - 4)$
$82 + 648$	$16 + 18(0)$
730	$34(0)$
	0

1. What error did Azalea make?

Azalea substituted 2 in for a , but did not use parentheses to show the multiplication of 8 and a . She wrote 82, which then looked like the number 82, instead of $8 \cdot 2$. She also substituted 6 in for b , but did not use parentheses to show the implied multiplication of $3b$. She wrote 36, which then looked like the number 36, instead of $3 \cdot 6$.

2. What error did Joshua make?

Joshua added $16 + 18$ before multiplying $18(0)$.



Error Analysis

Consider engaging students in a quick class discussion focusing specifically on the particular error each student made to highlight the common misconception. This will address student misunderstanding before they engage with the rest of the lesson.

11.2.2 Collaboration: Order of Operations

Component Overview: Students work together to number the order of operations in the correct sequence. Read the red boxes out loud to students, then have them work in partner pairs to find the numeric value of the given expression. After reviewing their work, engage the class in a discussion about the differences between their work and what is presented.



11.2.2 Collaboration: Order of Operations

1. Work with your partner to number each operation so that the steps for the order of operations are in the correct sequence.

3

Perform multiplication and division in order from left to right. (→)

2

Evaluate expressions with exponents.

4

Perform addition and subtraction in order from left to right. (→)

1

Perform operations inside grouping symbols.



An order of operations was first noted in First Year Algebra by Webster Wells and Walter Wilson Hart. The textbook was published in 1912.



Consider having students cut out the steps and glue them in the correct order on a reference poster to provide a kinesthetic entry point.



Challenge students to come up with a scenario in which they would go in a different order than what they chose. For example, when would you perform multiplication and division left to right before operations in grouping symbols?

"I would divide to simplify this fraction first before adding in the parentheses and still get the same value."

$$\frac{88}{22}(2 + 4)$$

11.2.2 Collaboration: Order of Operations (continued)

2. The problem below is an example from First Year Algebra.

EXAMPLE. Find the numerical value of the expression,

$$4ab + \frac{6c}{b} - d^3,$$

when $a=4$, $b=3$, $c=5$, $d=2$.

- a. What is the numerical value of the expression?
Show your work.

$$\begin{aligned} 4(4)(3) + \frac{6(5)}{3} - 2^3 \\ 48 + \frac{30}{3} - 8 \\ 48 + 10 - 8 \\ 50 \end{aligned}$$

- b. Check your work with your partner.

- c. The work shown in the textbook is below.

$$\begin{aligned} 4ab + \frac{6c}{b} - d^3 &= 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2 \cdot 2 \cdot 2 \\ &= 48 + 10 - 8 = 50. \end{aligned}$$

What is the difference between your work and the work from the book?

Emphasis should be placed on writing 2^3 as $2 \cdot 2 \cdot 2$. This is an illustration of how the order of operations was originally written without exponents. Instead, it was written out as a product.



An order of operations became more important when programmers needed a set of rules to program a computer to find values of numeric expressions. In 1883, Ada Lovelace described in her book about Charles Babbage's analytical engine, the first computer, that codes could be created so the device could handle letters and symbols along with numbers.



Notice and Wonder

Consider implementing a Notice and Wonder routine for question 2. Ask students to identify what they notice and wonder about their work and the work from the textbook.



What are some differences between numeric and algebraic expressions? The textbook changed d^3 to $2 \cdot 2 \cdot 2$. The order of operations in the textbook did not include exponents, so each exponent was rewritten as multiplication.

- When evaluating numeric expressions, how many values are possible?
- When evaluating algebraic expressions, how many values are possible?
- What information is needed in order to evaluate an algebraic expression?



Convey to students that the original computers were what is known today as calculators. Their only purpose was to do mathematical operations. Students who have seen *Hidden Figures* may be able to relate to this because the Katherine Johnson, Mary Jackson, and Dorothy Vaughn were referred to as computers. In the film, their jobs appeared to be threatened by the installment of computers which were machines that would do the calculations they were doing by hand.

11.2.3 Guided Instruction: Evaluating Expressions With Exponents

Component Overview: Guide students through the instructions at the beginning of this component. Engage students in a discussion about why evaluating exponents should be done before other multiplication and division in the expression. Then have students work independently to evaluate the numeric expressions. Finally, engage students in a discussion to debrief the questions and consolidate their understandings.



11.2.3 Guided Instruction: Evaluating Expressions with Exponents

The work in the book, First Year Algebra, showed 2^3 written out as multiplication. This method can get complex as shown in the table.

$(-3)^3 + 2^6 \div -16 + 5$
<i>Exponents written as multiplication</i>
$((-3)(-3)(-3)) + (2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) \div -16 + 5$

$(-3)^3 + 2^6 \div -16 + 5$
<i>Evaluate expressions with exponents first</i>
$-27 + 64 \div -16 + 5$

- When writing $(-3)^3$ and 2^6 using multiplication, both were written inside a set of parentheses.
- In order of operations, parentheses indicate that the operations inside should be done **before** other steps.
- Therefore, evaluating exponents is done before other multiplication and division when following the order of operations.

- Evaluate $(-3)^3 + 2^6 \div -16 + 5$.

$$\begin{aligned} &-27 + 64 \div -16 + 5 \\ &-27 + (-4) + 5 \\ &-31 + 5 = -26 \end{aligned}$$

- Evaluate each expression, then determine if the expressions are equivalent.

Expression A	Expression B	Equivalent?
$(-3)^4 = 81$	$-(3^4) = -81$	No
$-(9)^2 = -81$	$(-9)^2 = 81$	No
$(-5^2) = -25$	$(-5)^2 = 25$	No
$26 - 4^3$ $= 26 - 64$	$26 - (-4)^3$ $= 26 + (64)$	No
$17 + (-2^4)$ $= 17 + (-16)$	$17 - (2^4)$ $= 17 - 16$	Yes



Students may need a brief review of evaluating exponents, particularly with negative bases. The second example in 11.2.3 Guided Instruction provides an opportunity to monitor to assess formatively how the students approach this within the order of operations. Address the misconceptions as needed with specific students or the whole class through modeling.



This activity provides an entry point for visual learners through the use of color-coding the expressions in the table.

11.2.3 Guided Instruction: Evaluating Expressions With Exponents (continued)

3. Describe a strategy that can be used when evaluating expressions with exponents that include a negative sign or a subtraction sign.

Answers will vary. Guide students to even/odd exponents, the placement of a negative in front of the parentheses as opposed to inside.

4. Elizabeth evaluated $b + a^3 \cdot 7 - 9$ when $a = -4$ and $b = 17$. Her work is below.

- a. Circle Elizabeth's error.

$$17 + (-4)^3 \cdot 7 - 9$$

$$17 + (-64) \cdot 7 - 9$$

$$(-47) \cdot 7 - 9$$

$$-329 - 9$$

$$-338$$

- b. Which step of the order of operations did she not apply correctly?
- Ⓐ Evaluate expressions with exponents.
 - Ⓑ Perform operations inside grouping symbols.
 - Ⓒ Perform multiplication and division in order from left to right.
 - Ⓓ Perform addition and subtraction in order from left to right.



The order of performing computations is first to work within grouping symbols using the **order of operations**. Then, simplify terms with exponents. Next, while reading from left to right, perform multiplication and division in the order in which it appears. Finally, while reading from left to right, perform addition and subtraction in the order in which it appears.



Solidify understanding by asking these questions:

- What are some differences between numeric and algebraic expressions? (Numeric expressions contain all known values while algebraic expressions include some unknown value(s).)
- When evaluating numeric expressions, how many answers are possible? (Only one.)
- What information is needed in order to “evaluate” an algebraic expression? (Specific values for the variable(s).)



While reviewing the order of operations in the vocabulary box, remind students that up to this point it has been shown that these are not hard and fast rules but should be used to guide them when they determine a value. Although there are some things where the order does not matter there are others where the order does matter. It may be helpful to create an anchor chart or reference cards with examples for students to refer to as they build their understanding.

11.2.4 Try It: Discovering the Importance of Left to Right

Component Overview: Students work independently to evaluate expressions in different ways, working to determine how changing grouping symbols or the order in which operations are performed changes the value of the expression. After discussing question 1, use your discretion to determine if students can work independently or would do better checking in with a partner regularly.



11.2.4 Try It: Discovering the Importance of Left to Right

1. Mia is evaluating the expression, $x^3 \div r \cdot (k^2 - 5)$ for $k = 3$, $r = -8$ and $x = 4$. Her first few steps are shown:

$$\begin{aligned} 4^3 \div -8 \cdot (3^2 - 5) \\ 64 \div -8 \cdot (9 - 5) \\ 64 \div -8 \cdot 4 \end{aligned}$$

- a. Mia thinks that she could finish her work by either finding $64 \div -8$ or $-8 \cdot 4$ first. Show the work for both possibilities to determine if Mia is correct.

$64 \div -8 \cdot 4$	$64 \div -8 \cdot 4$
$(64 \div -8) \cdot 4$	$64 \div (-8 \cdot 4)$
$-8 \cdot 4 = -32$	$64 \div -32$
	-2

The answers are not the same.

- b. Was Mia correct when she stated that both methods would work? Use the steps of the order of operations to justify whether or not Mia is correct.

Multiplication and division should be done from left to right.

2. Answer the following problems.

- a. Evaluate $r \cdot W \div x \div m$ when $m = 2$, $r = -4$, $W = -60$ and $x = -10$. Show your work.

$$\begin{aligned} -4 \cdot (-60) \div -10 \div 2 \\ 240 \div -10 \div 2 \\ -24 \div 2 \\ -12 \end{aligned}$$



Compare and Connect

Each question in this section provides an opportunity for students to compare how changing the order in which operations are performed changes the value of the expression. Consider modeling a Think Aloud for question 1b to model the thinking students should be doing. Then have students engage in this Compare and Connect with a partner before completing question c for 2 and 3.



Consider splitting the class into differentiated groups to complete this section after modeling the Think Aloud to answer question 1. Guide students who may need more support through the thinking for question c.

11.2.4 Try It: Discovering the Importance of Left to Right (continued)

- b. Evaluate $(r \cdot W) \div (x \div m)$ when $m = 2$, $r = -4$, $W = -60$ and $x = -10$. Show your work.

$$\begin{aligned} &(-4 \cdot -60) \div (-10 \div 2) \\ &240 \div -5 \\ &-48 \end{aligned}$$

- c. How did the use of parentheses change the answers?

The parentheses changed the order so that $-10 \div 2$ was divided first instead of $-240 \div -10$ being first.

The parentheses around $r \cdot W$ did not change what was done without parentheses.

3. Answer the following problems.

- a. Evaluate $d \div k \cdot y \div B$ when $B = 3$, $d = -72$, $k = -6$ and $y = 9$. Show your work.

$$\begin{aligned} &-72 \div -6 \cdot 9 \div 3 \\ &12 \cdot 9 \div 3 \\ &108 \div 3 \\ &36 \end{aligned}$$

- b. Evaluate $d \cdot k \div y \div B$ when $B = 3$, $d = -72$, $k = -6$ and $y = 9$. Show your work.

$$\begin{aligned} &-72 \cdot -6 \div 9 \div 3 \\ &432 \div 9 \div 3 \\ &48 \div 3 \\ &16 \end{aligned}$$

- c. How did rearranging the operations change the answers?

Answers will vary. By changing the order of operations going from left to right meant that $d \div k$ was changed to $d \cdot k$ which gave a different value.



- What happens when you use parentheses in the expression? How do they change the value of the expression? Why does it change the value?
- How does the value of the expression change when you rearrange the order of the operations?

11.2.5 Your Turn!

Component Overview: Students determine which algebraic expressions are equal by substituting the same two values in for the two variables in each of the three variable expressions. Students demonstrate their understanding of the order of operations by evaluating the expressions.



11.2.5 Your Turn!

1. Answer the following problems.

- a. Circle the expressions you think will result in the same value.

$$7 - (y - 11)^2 + 6V \quad 7 - y^2 - 11^2 + 6V \quad 7 - y^2 + 11^2 + 6V$$

- b. Find the value of $7 - (y - 11)^2 + 6V$ when $V = -9$ and $y = 4$.

$$\begin{aligned} 7 - (4 - 11)^2 + 6(-9) \\ 7 - (-7)^2 + 6(-9) \\ 7 - 49 + (-54) \\ -42 + -54 = -96 \end{aligned}$$

- c. Find the value of $7 - y^2 - 11^2 + 6V$ when $V = -9$ and $y = 4$.

$$\begin{aligned} 7 - 4^2 - 11^2 + 6(-9) \\ 7 - 16 - 121 + (-54) \\ -9 - 121 + -54 \\ -130 + -54 = -184 \end{aligned}$$

- d. Find the value of $7 - y^2 + 11^2 + 6V$ when $V = -9$ and $y = 4$.

$$\begin{aligned} 7 - (4)^2 + 11^2 + 6(-9) \\ 7 - 16 + 121 + (-54) \\ -9 + 121 + (-54) \\ 112 + -54 = 58 \end{aligned}$$

- e. Which expressions, if any, resulted in the same value?

None



Visual learners may benefit from using different colored pencils to substitute each variable. This can help keep track of their work and encourage precision in evaluation.



Challenge students to find a variable pair that will result in the expressions all having the same value. Encourage them to explain why the expressions do not and cannot have the same value using the rules of the order of operations.

11.2.6 Wrap-Up: Error Analysis

Component Overview: In this formative assessment, students demonstrate their understanding of evaluating algebraic expressions. Students analyze two students' work to identify errors. Then they explain how to fix the errors in a note to the students.



11.2.6 Wrap-Up: Error Analysis

1. Nadine and Nora evaluated the expression $r^3 \div x \cdot c$ for $c = 9$, $r = -6$ and $x = -3$. Both students made errors in their work shown. Circle their errors.

Nadine's Work	Nora's Work
$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-216 \div (-3) \cdot 9$ $-216 \div -27$ 8	$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-18 \div (-3) \cdot 9$ $6 \cdot 9$ 54



Consider allowing students to record their answers using an app like FlipGrid. This will be beneficial for students who may understand the content but struggle with written response.

2. Write a short explanation to each student to explain the mistake and what she needs to remember about evaluating expressions in the future.

To Nadine: Nadine, The order of operations states to multiply and divide from left to right. In the step $-216 \div (-3) \cdot 9$ you should divide $-216 \div (-3)$ then multiply $72 \cdot 9$.	To Nora: Nora, $(-6)^3$ is a shortcut way of writing $-6 \cdot -6 \cdot -6$, where: $-6 \cdot -6 = 36$ $36 \cdot -6 = -216$
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